



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

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THE PROBLEM: INVISIBLE RIPPLE EFFECTS ON THE LINE

A vehicle assembly line is a chain of connected stations. A slowdown at one station creates queues and cuts output. A defect introduced early can pass through many vehicles before it's caught, causing rework and quality loss.

Bottleneck Formation

When a station slows down, it cannot keep up with the incoming vehicles. This causes vehicles to build up behind it, while the stations after it may not have enough work, reducing the overall output of the line.

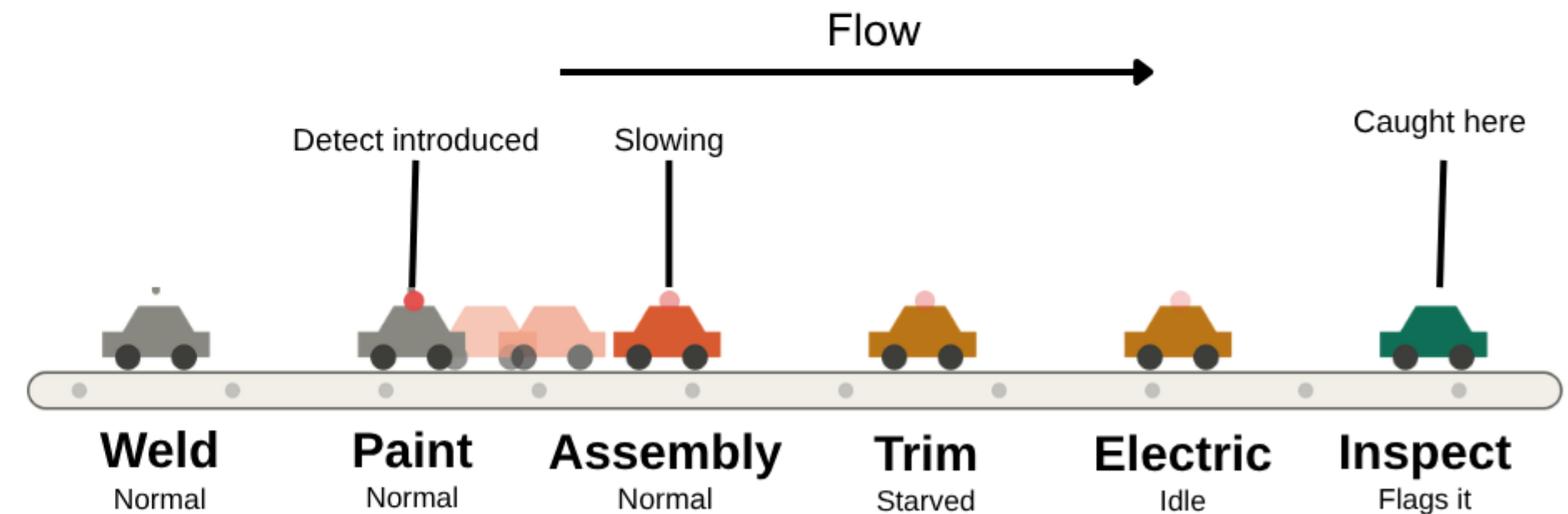
Defect Propagation

If a process drifts out of range, the same defect can repeat across many vehicles before inspection catches it.

Limited Sensor Data

Some stations lack sensors, so missing or unreliable data makes it hard to see the line's true state and find where problems start.

\$22K/min	Average reported cost of stopped production
\$2M+/hr	Reported cost of one hour of automotive-line downtime
70%+	Automotive manufacturers deploying Digital Twins



“By the time you spot the problem, the damage is already done.”

TWINLINE.AI: BUILDING A DIGITAL TWIN OF THE LINE

TWINLINE.AI tracks the production line, spots bottlenecks and defects early, and helps operators test and choose the right action.

SENSE

Production data • Cycle time •
Queue • Machine health

UNDERSTAND

Line state • Station status •
Flow

PREDICT

Bottleneck risk • Defect risk
What happens next?

CONFIDENCE

High • Medium • Low

SIMULATE

Test actions and compare
options

RECOMMEND

Recommend the best action
based on the predicted outcome

ACT

AI recommends.
Humans decide.

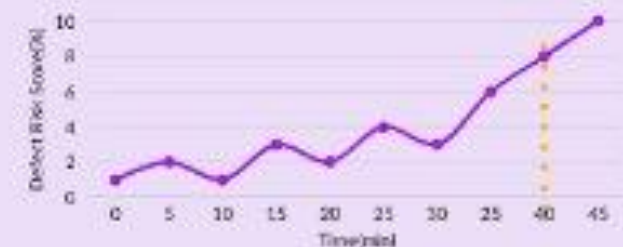
PREDICTION ENGINE

BOTTLENECK PREDICTION

Queue growth + utilization drift
->Target: 15–20 min early warning



Illustrative: Bottleneck Prediction Trends



DEFECT PREDICTION

Sensor-pattern drift
-> Defect risk before inspection

SEE THE RESULT BEFORE YOU ACT

PREDICT -> WHAT-IF -> SIMULATE -> COMPARE -> RECOMMEND
Example: Move one operator to Station 1 (Illustrative what-if scenario)

	BASELINE	WHAT-IF
Throughput	950	1,045
Max queue	38	24
Utilization	62%	74%

DATA CONFIDENCE

DIRECT DATA

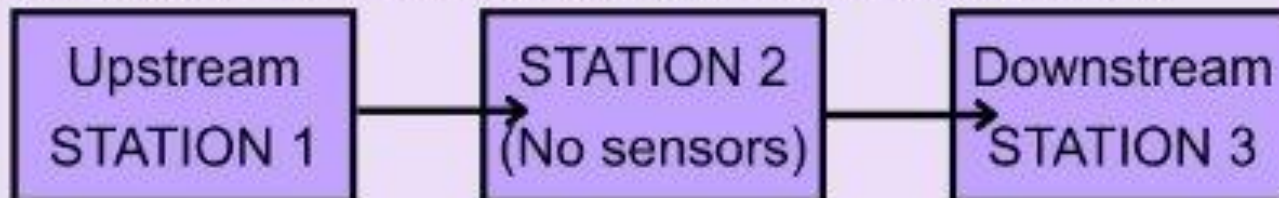
Sensors + history = HIGH

PARTIAL DATA

Some signals + history = MEDIUM

NO DIRECT SENSOR

Nearby station data + process signals → LOW CONFIDENCE



Inference = verification before intervention

NO FALSE CERTAINTY

Insufficient data triggers verification

References:

- Mendi (2022)
- Nandy et al. (2026)
- Azilen-GenAI & Digital Twin
- Other supporting sources

IMPACT

Earlier bottleneck
detection

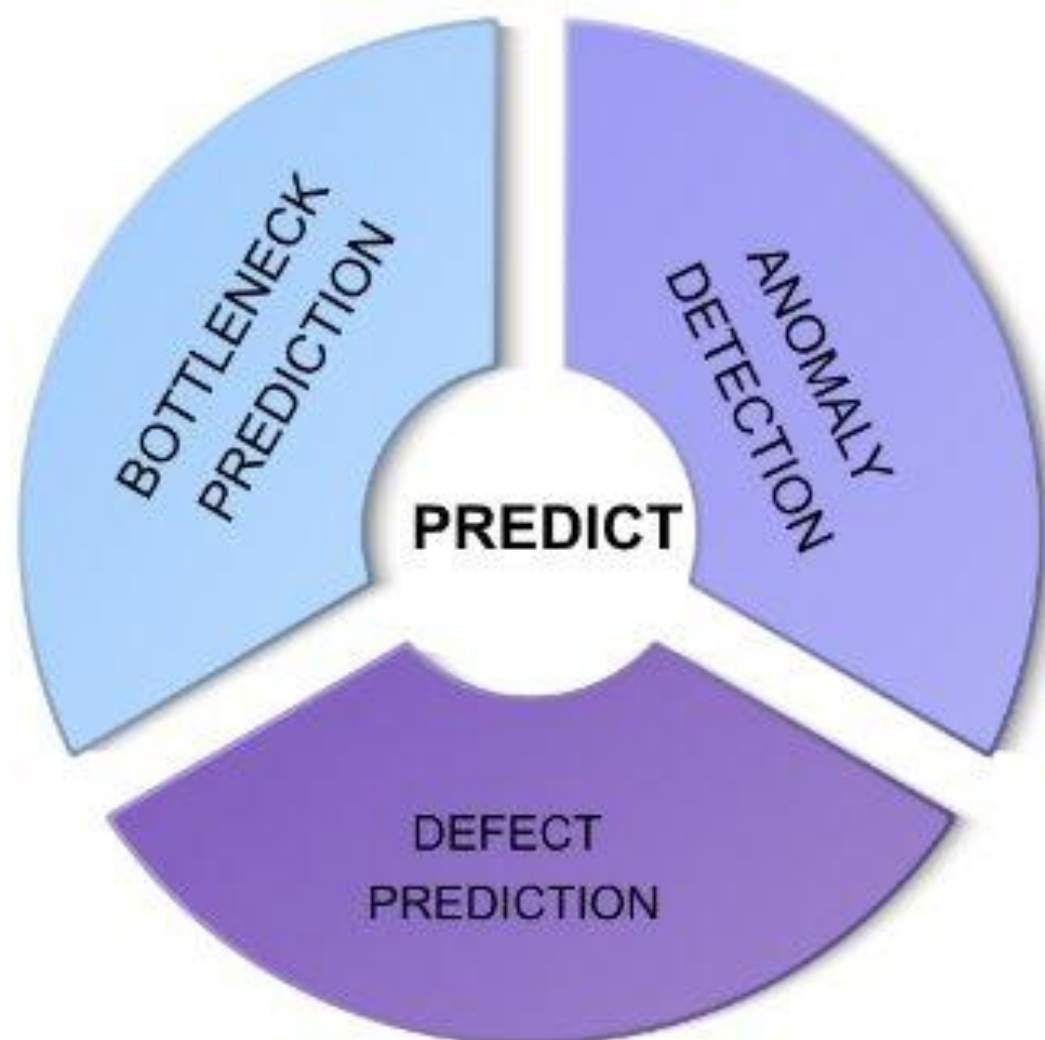
Fewer unplanned disruptions

Higher Quality
output

Lower Operating Cost

Smarter, Faster decisions by
operators

PREDICTION ENGINE



BOTTLENECK PREDICTION

Signals

- Queue growth
- Cycle-time drift
- Utilization

Output

Risk + Lead Time

DEFECT PREDICTION

Signals

- Torque
- Vibration
- Temperature
- Part quality

Output

Defect Risk

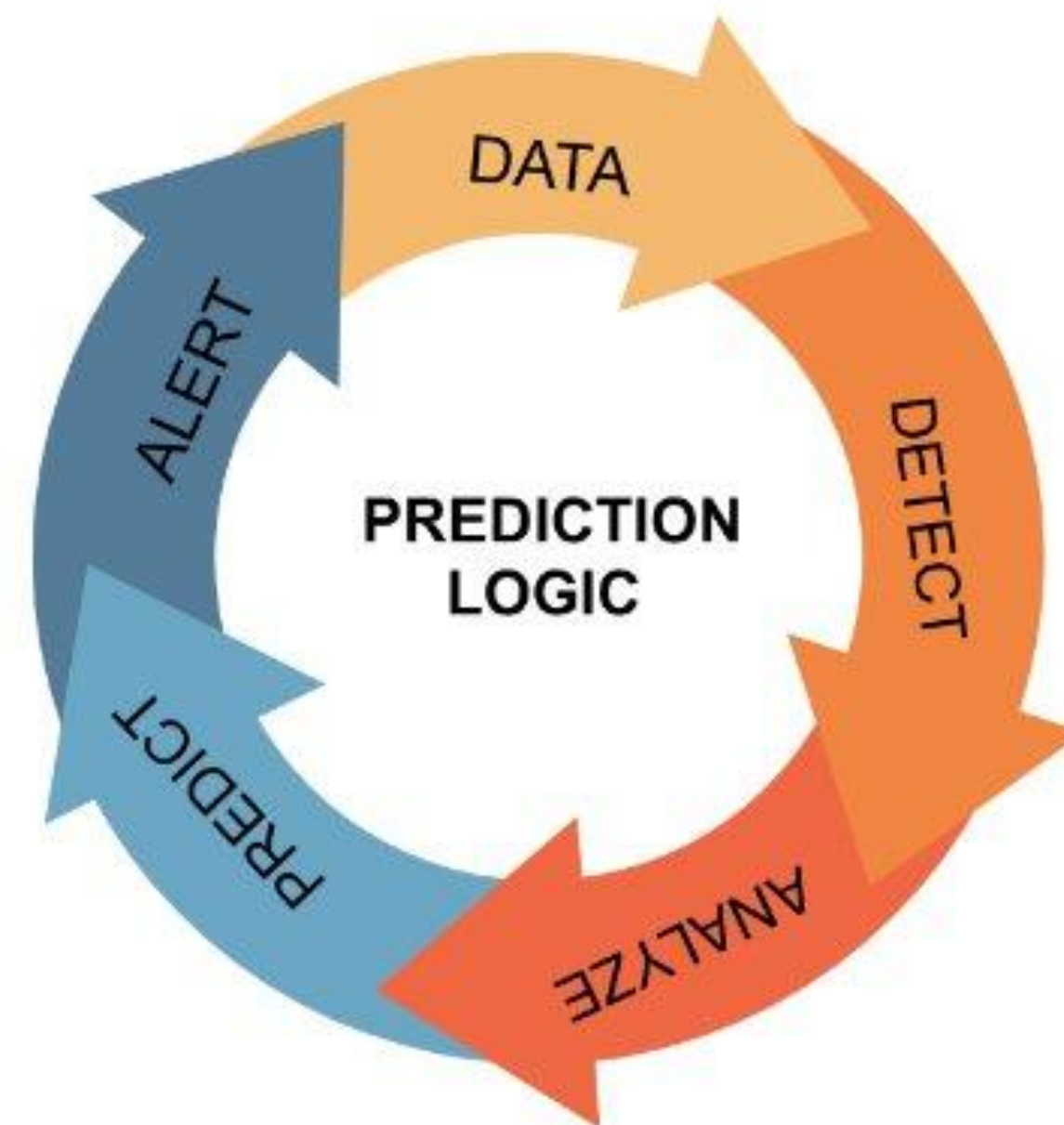
ANOMALY DETECTION

Signals

- Process baseline
- Sensor drift
- Historical deviation

Output

Early Warning



SENSOR-POOR STATION

UPSTREAM + DOWNSTREAM + HISTORY



INFERRED
CONDITION

LOWER CONFIDENCE!!!

CONFIDENCE ENGINE

SENSOR COVERAGE

Direct vs inferred data

DATA RECENCY

Fresh vs stale signals

HISTORICAL STABILITY

Normal behaviour baseline

SIGNAL AGREEMENT

Multiple signals support prediction

CONFIDENCE SCORE

Evidence strength

Example Output

Station 14

Bottleneck Risk: 78%

Confidence: MEDIUM

→ Verify before action
That's enough.

PREDICTION CONFIDENCE

HIGH

Direct data

MEDIUM

Partial sensor data based
on historical trends

LOW

Limited direct data from
inferred signals

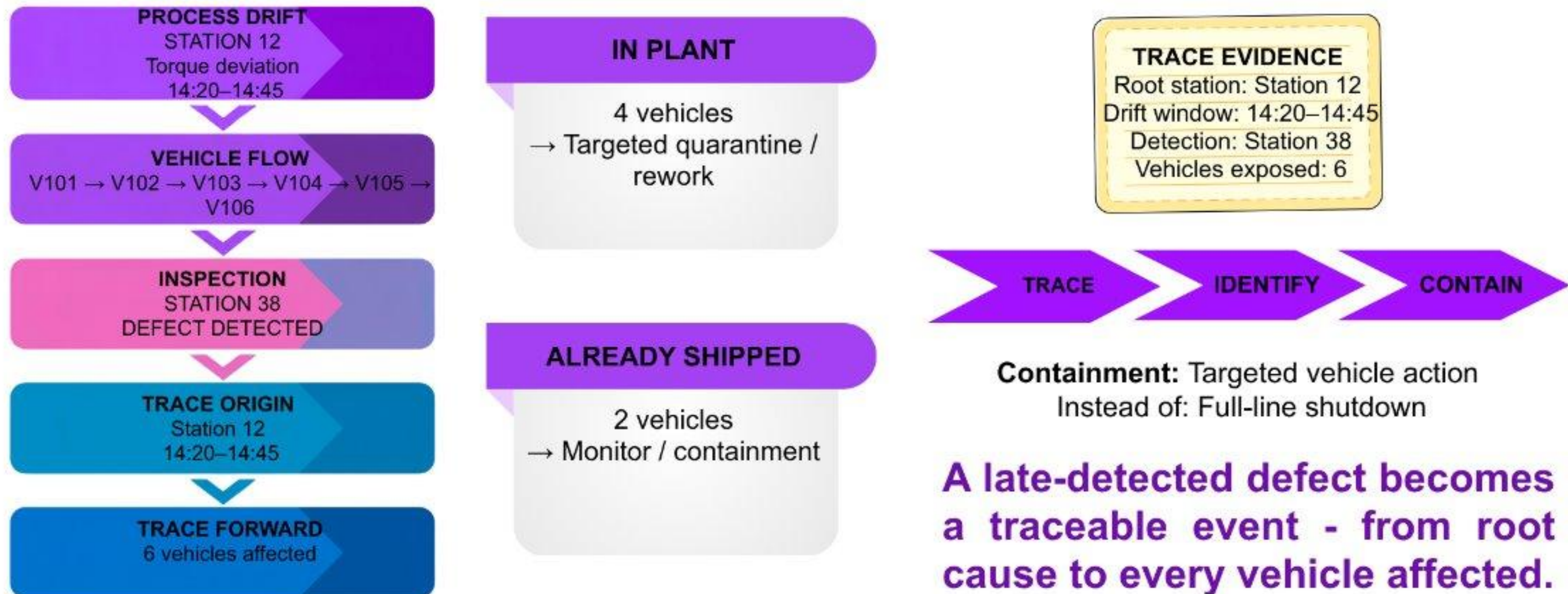
ABSTAIN

Insufficient data, need to verify first

DEFECT GENEALOGY

TRACE THE DEFECT. FIND EVERY VEHICLE IT TOUCHED.

When a defect is detected late, TwinLine.AI traces it backward to its likely origin and forward to the vehicles exposed to it.

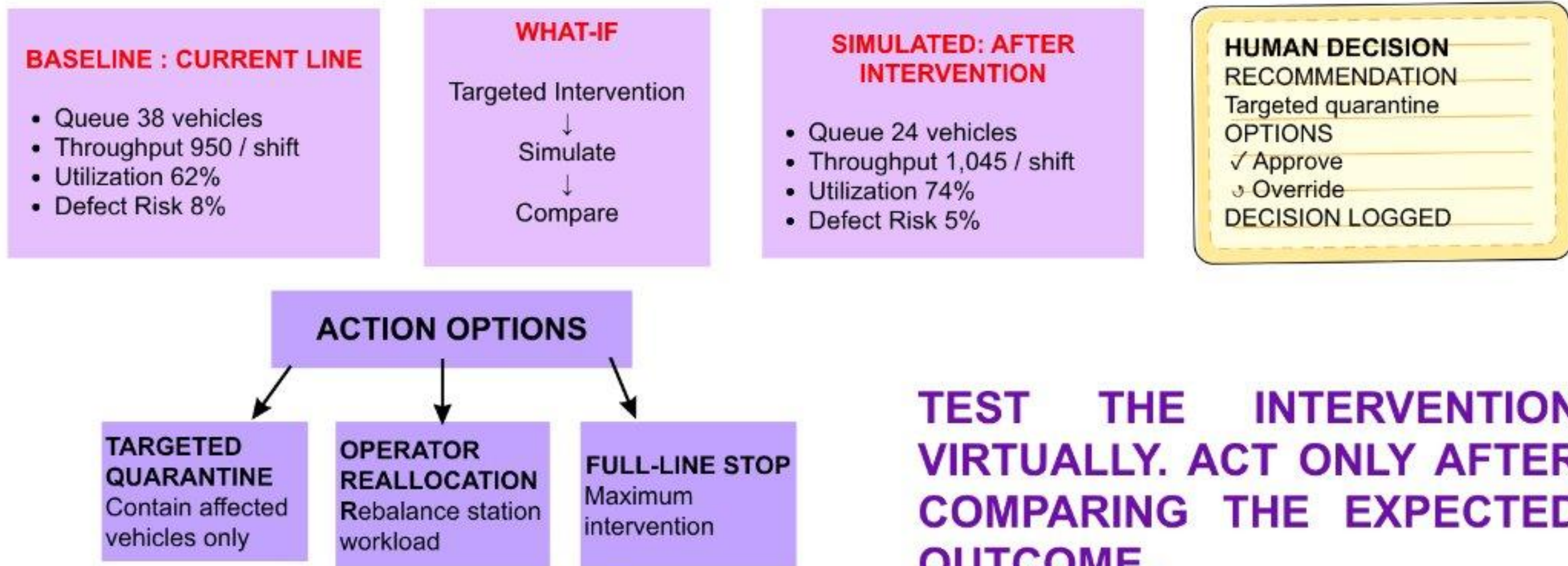


WHAT-IF SIMULATION

SEE THE RESULT BEFORE YOU ACT

TwinLine.AI tests possible interventions against the current line state before a human takes action.

Illustrative Prototype Output.



TEST THE INTERVENTION VIRTUALLY. ACT ONLY AFTER COMPARING THE EXPECTED OUTCOME.

SIMULATE → COMPARE → RECOMMEND

OT DEPLOYMENT

DEPLOYMENT PRINCIPLES

READ-ONLY

- No live PLC changes
- Existing controls remain untouched.

PHASED

- Maintenance-window upgrades
- Add sensors during scheduled maintenance.

HUMAN-LED

- No autonomous control
- Recommendations require verification.

CRUX

We can deploy TwinLine.AI around the existing factory without changing live PLC control, improving decision making without interrupting the line.

OT SAFETY BY DESIGN

- Read-only data access
- Network segmentation
- Least-privilege access
- Human approval before action

EXISTING PLANT

PLC / SCADA Sensors
Historian / MES
Manual Checks

READ-ONLY DATA LAYER

Collect
Normalize
Timestamp

TWINLINE.AI

Digital Twin
SPC + ML
Confidence

DECISION LAYER

Predict
Trace
Simulate

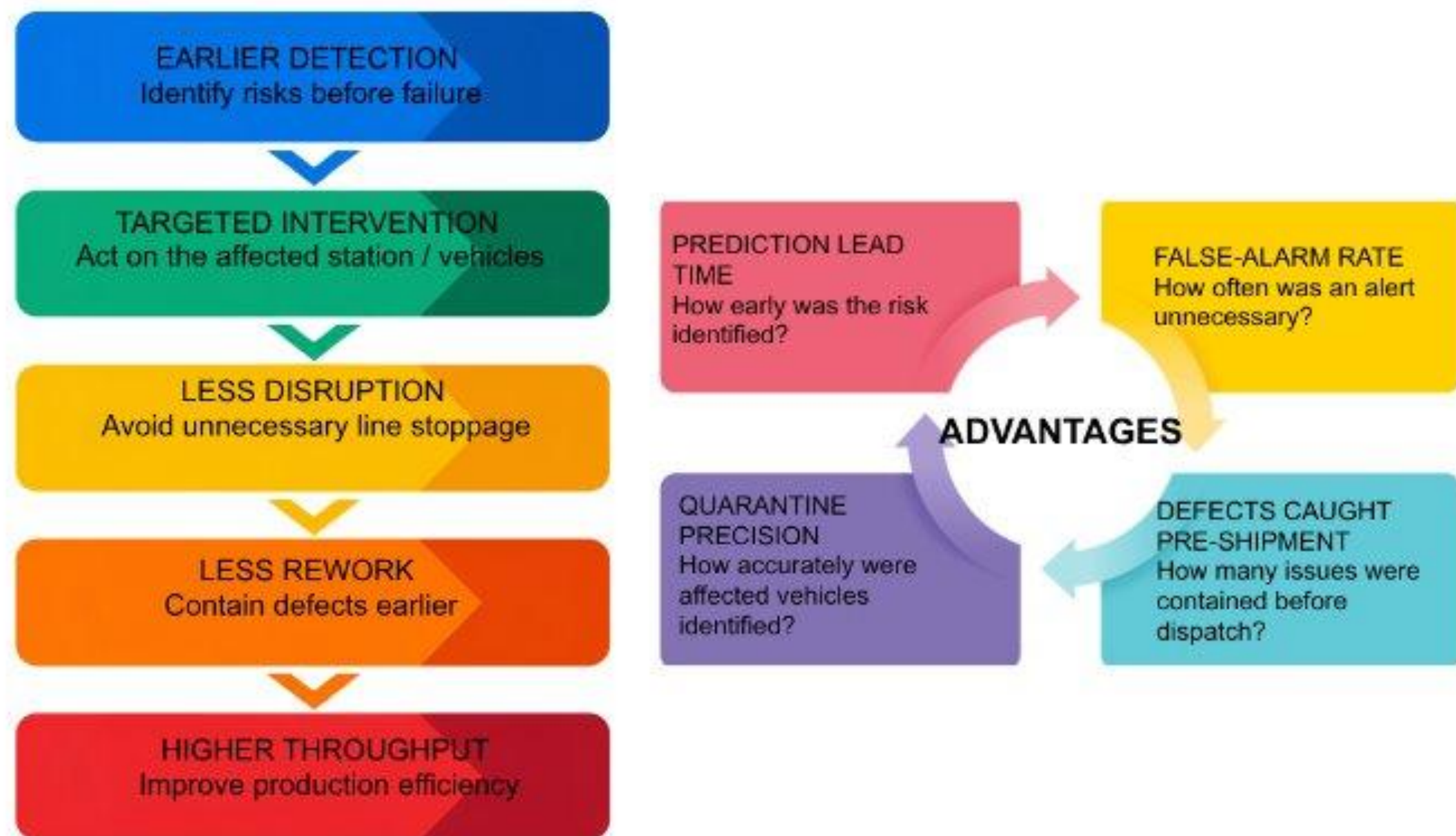
HUMAN

Verify
Approve / Override
Act

BUSINESS IMPACT

FROM PREDICTION TO BUSINESS VALUE

TwinLine.AI converts early warnings into measurable operational improvements.



CURRENT STATE	TWINLINE.AI
Late detection → Broad investigation → Larger disruption → Higher rework	Early prediction → Vehicle-level trace → Targeted containment → Lower disruption

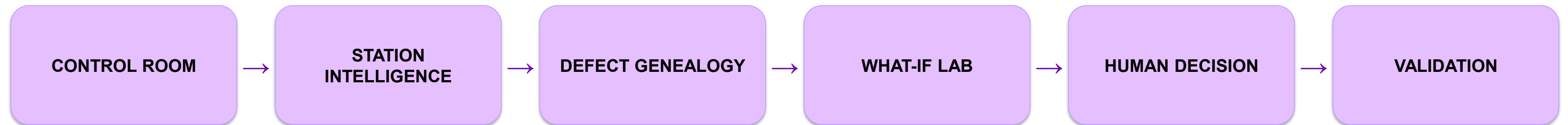
The value is not just predicting a problem - it is reducing the operational impact of that problem.

Note:Business impact figures to be populated from validated prototype results; financial estimates are illustrative assumptions.

INSIDE TWINLINE.AI : THE LIVE PROTOTYPE

WHAT YOU'RE ABOUT TO SEE IS LIVE, NOT SCRIPTED

Every screen ahead reflects the same 40-station assembly line simulation, running interactively.



KEY HIGHLIGHTS

- Plant-wide Control Room monitoring
- Station-level risk & confidence intelligence
- Defect genealogy and affected-vehicle tracing
- What-if intervention simulation
- Human approve/override decision logging
- Validation & performance metrics

INSIDE TWINLINE.AI : KEY FEATURES

CONTROL ROOM

See the entire production line at a glance

- Monitor all 40 stations
- View station risk and alert status
- Detect emerging process abnormalities
- See sensor/evidence coverage
- Navigate directly to stations requiring attention

STATION INTELLIGENCE

Understand why a station is being flagged

- Station-level defect & bottleneck risk
- Process signal trends
- SPC / statistical evidence
- Sensor coverage and data quality
- Confidence score based on available evidence

DEFECT GENEALOGY

Trace a downstream defect back through the process

- Follow vehicle history through stations
- Identify the **likely origin station**
- Rank the origin using observed signal onset + process sequence
- Identify potentially affected vehicles
- Separate in-plant containment from already-shipped units

WHAT-IF LAB

Test the decision before taking action

- Continue production
- Repair the suspected station
- Targeted quarantine
- Full-line stop
- Compare projected queue, residual risk and disruption

HUMAN DECISION

Keep the operator in control

- Review the model recommendation
- **Approve** the recommended action
- **Override** when operational judgement differs
- Add a decision note
- Maintain an auditable decision log

VALIDATION & ROI

Measure whether the approach works

- Precision / Recall / F1
- ROC-AUC
- False-positive rate
- Detection lead time
- Synthetic backtest disclosure
- Illustrative ROI / minutes saved

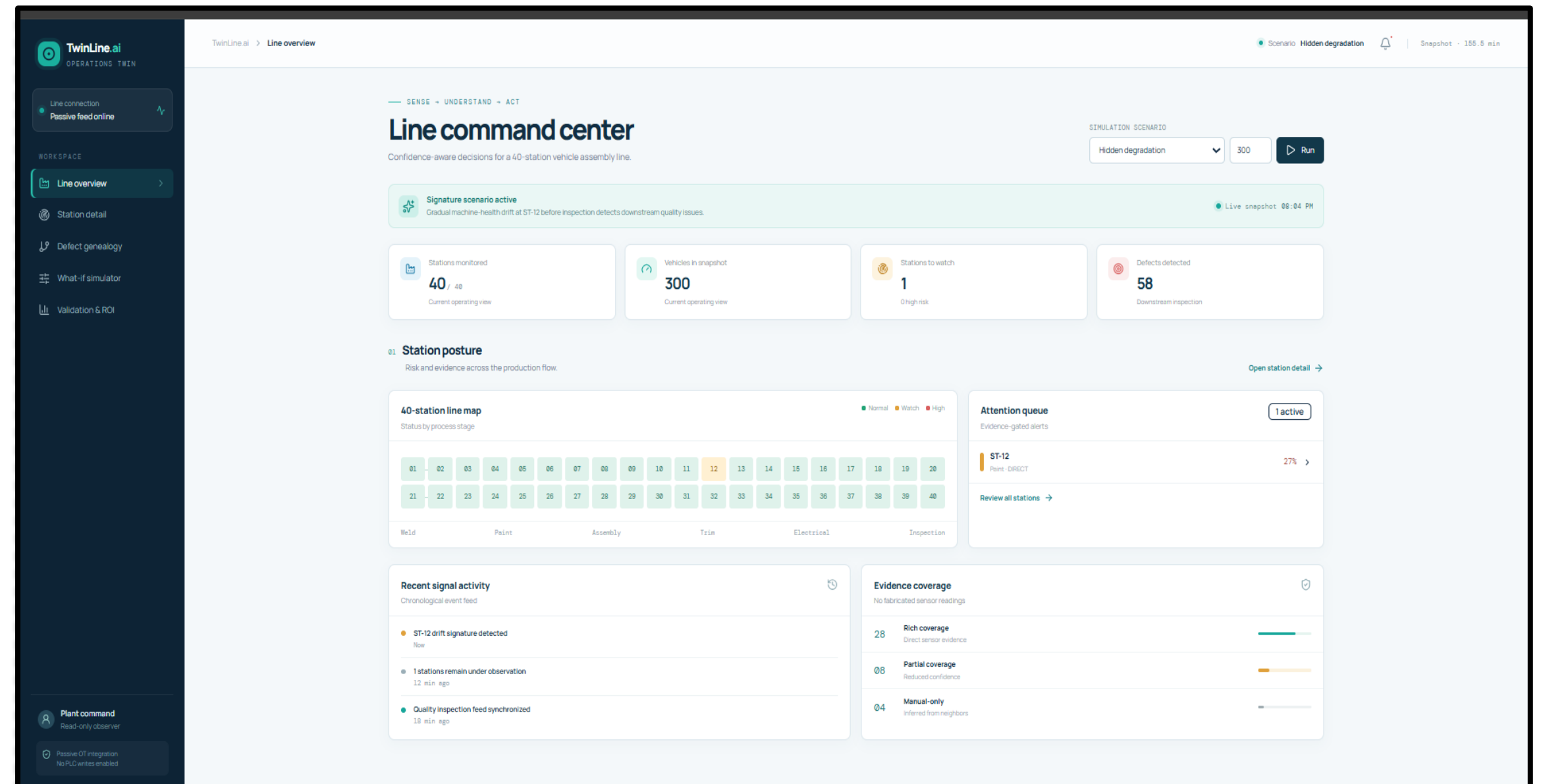
TwinLine.ai turns raw process signals → evidence → likely origin → affected vehicles → intervention simulation → human-approved action.

CONTROL ROOM : PLANT-WIDE VISIBILITY

ONE SCREEN, 40 STATIONS, FULL CONTEXT

The first screen an engineer sees - live risk posture across the entire line.

- **Stations monitored:** 40 of 40 tracked live
- **Vehicles in snapshot:** 300
- **Stations to watch:** flagged automatically, not manually
- **Evidence coverage:** shown per station - rich, partial, or manual-only
- **Signature scenario banner:** explains what condition is currently active

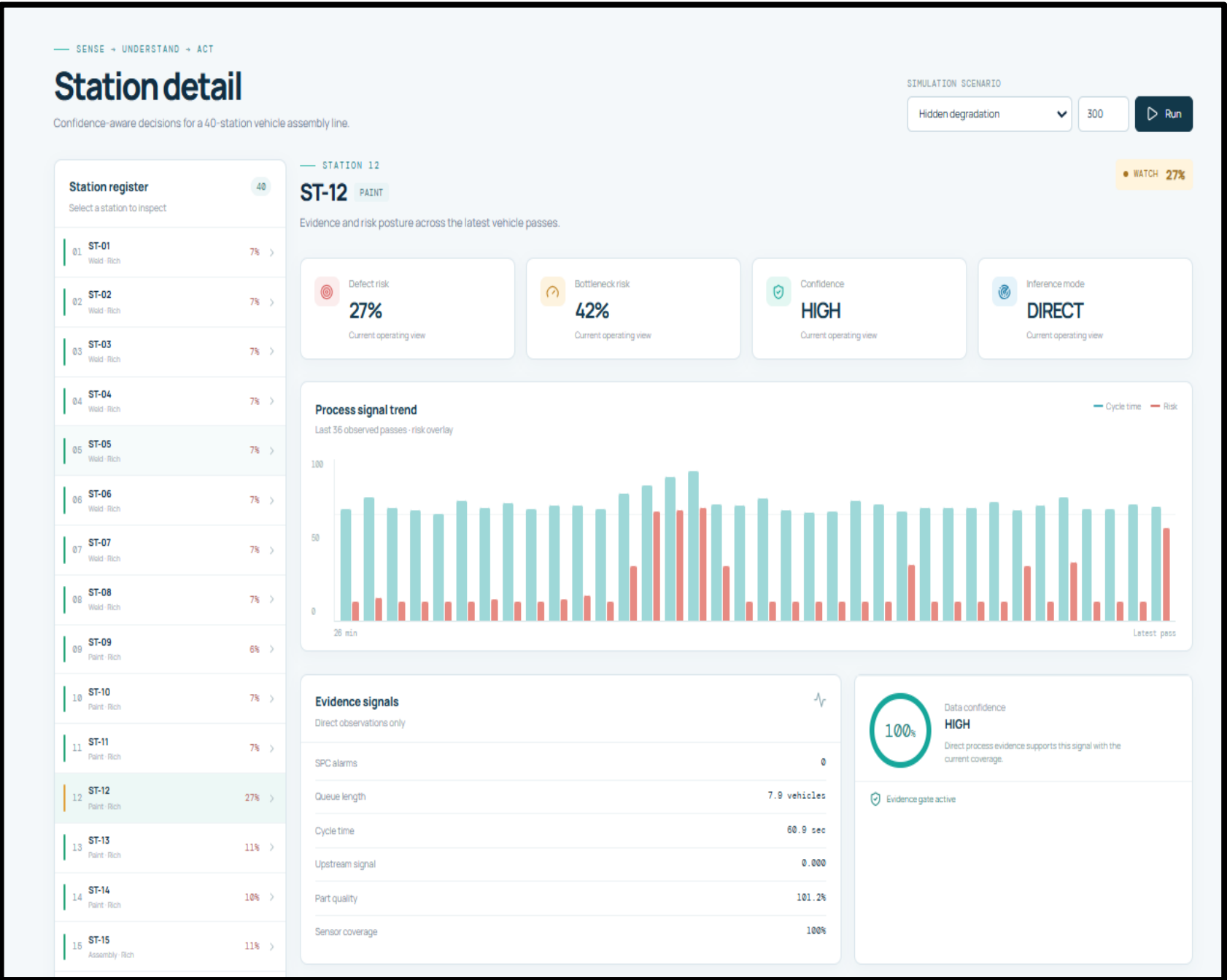
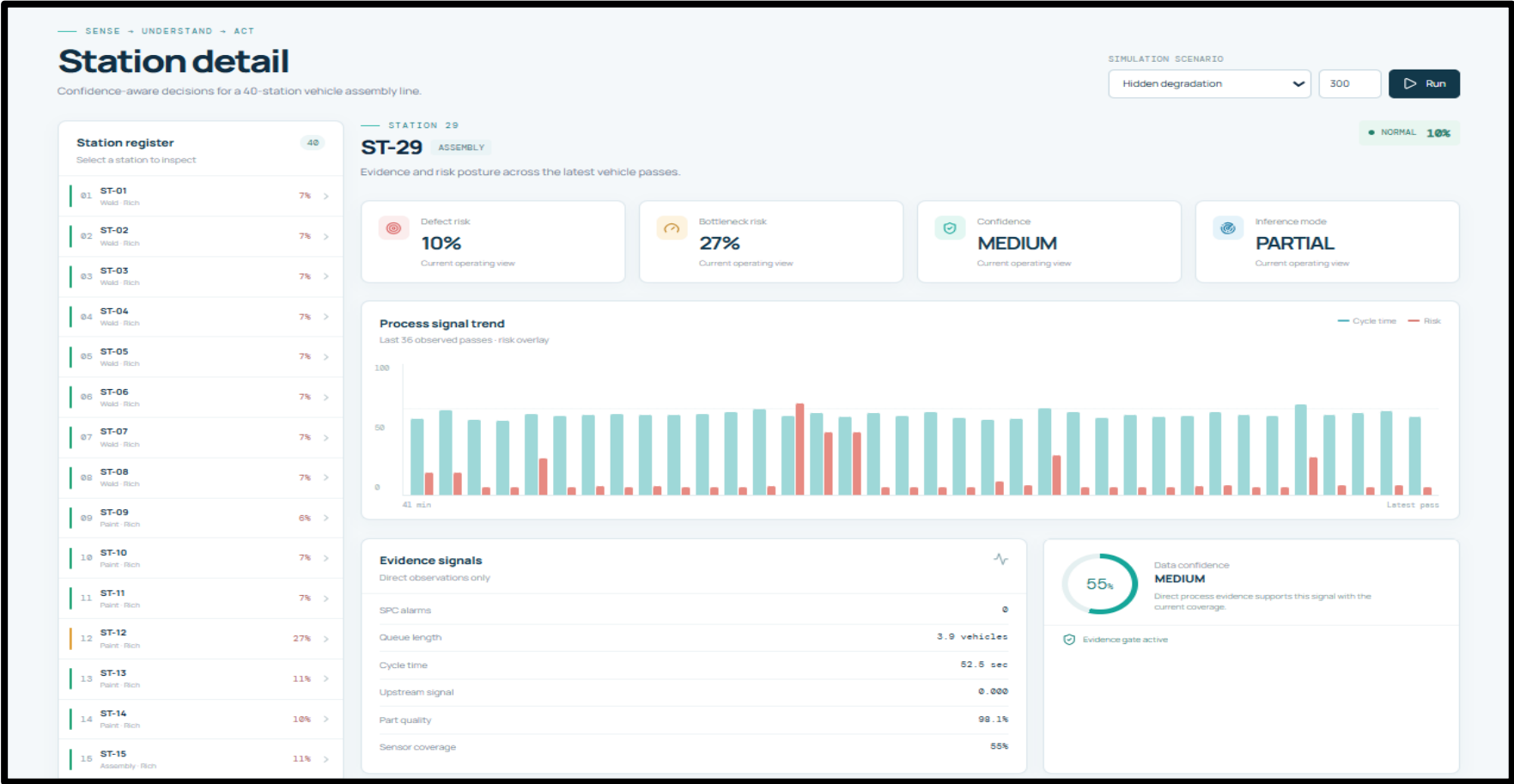


STATION INTELLIGENCE : EVIDENCE BEHIND THE RISK

DRILL DOWN FROM PLANT TO STATION

Selected: ST-12 (Paint) - every risk score comes with the evidence behind it.

- **Defect risk:** 27%
- **Bottleneck risk:** 42%
- **Confidence:** HIGH
- **Signals tracked:** queue length, cycle time, upstream signal, part quality, sensor coverage
- **Process trend:** risk overlay across last 36 observed passes

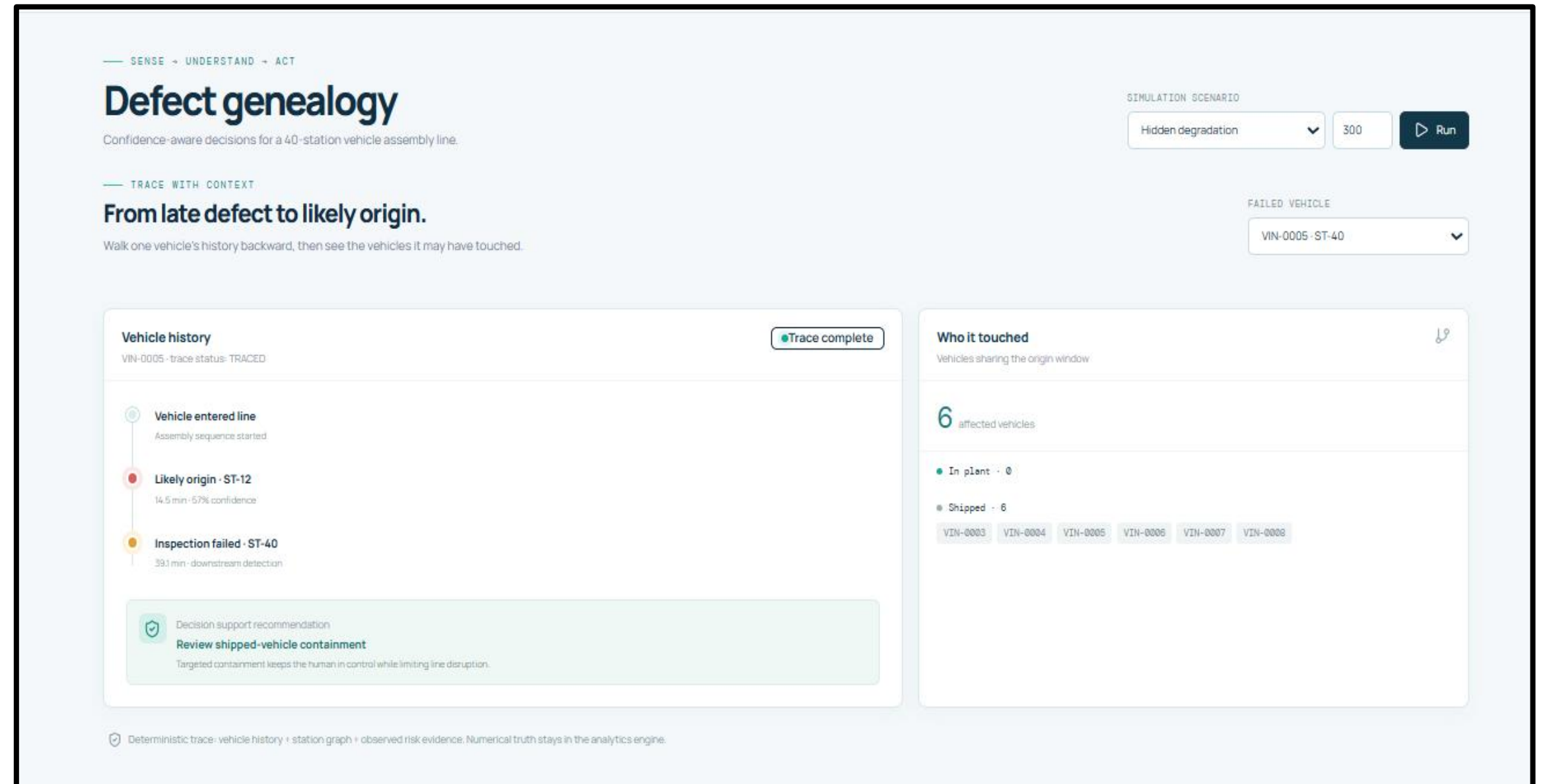


DEFECT GENEALOGY : TRACED LIVE

FROM LATE DEFECT TO LIKELY ORIGIN

TwinLine.AI ranks candidate defect origins using process-sequence and signal-onset evidence.

- **Vehicle traced:** VIN-0005, backward through process history
- **Likely origin:** ST-12, 57% confidence
- **Downstream detection:** inspection failed at ST-40
- **Affected vehicles:** 6 share the same origin window
- **Trace method:** deterministic - vehicle history + station graph + observed evidence



WHAT-IF LAB : SIMULATE BEFORE YOU ACT

COMPARE INTERVENTIONS, NOT GUESSES

Model-backed comparison of intervention paths before anything touches the real line.

- **Response paths:** Continue line, Repair station, Targeted quarantine, Full line stop
- **Example run:** Continue at ST-12
- **Projected outcome:** 7.2 vehicles expected queue, 27% residual risk, Low disruption
- **Comparison:** before vs. simulated-after, side by side

The WHAT-IF simulator consists of three sections :

- Intervention setup
- Human decision
- Decision log

The screenshot displays the 'What-if simulator' interface, which is designed for confidence-aware decisions for a 40-station vehicle assembly line. The interface is divided into three main sections: Intervention setup, Human decision, and Decision log.

Intervention setup: This section allows users to model from the current snapshot. It features a 'RESPONSE PATH' with four options: 'Continue line' (Accept current trajectory), 'Repair station' (Reduce source risk), 'Targeted quarantine' (Contain affected vehicles), and 'Full line stop' (Maximum risk reduction). The 'Targeted quarantine' option is currently selected. Below this, the 'TARGET STATION' is set to 'ST-12 - Paint'. A 'Compare outcome' button is located at the bottom of this section.

Human decision: This section is for recording the accountable action. It includes two buttons: 'Approve recommendation' (highlighted in green) and 'Override recommendation'. Below these buttons is a text input field for 'Optional decision note' and a 'Log decision' button.

Decision log: This section shows the decision log for the current session. It currently displays 'No decisions logged yet.'

At the top right, there is a 'SIMULATION SCENARIO' dropdown menu set to 'Hidden degradation', a '300' value, and a 'Run' button. A 'Human approval required' notification is also present.

WHAT-IF LAB : SIMULATE BEFORE YOU ACT

SENSE → UNDERSTAND → ACT

What-if simulator

Confidence-aware decisions for a 40-station vehicle assembly line.

DECISION COCKPIT

Simulate before you act.

Compare disruption, queue pressure, and residual risk—then log the human decision.

SIMULATION SCENARIO

Hidden degradation

300

Run

Human approval required

Intervention setup

Modelled from the current snapshot

RESPONSE PATH

Continue line

Accept current trajectory

Repair station

Reduce source risk

Targeted quarantine

Contain affected vehicles

Full line stop

Maximum risk reduction

TARGET STATION

ST-12 · Paint

Compare outcome

Projected outcome - ST-12

Continue

LOW disruption

Expected queue	7.2 vehicles	Residual risk	27%
Queue impact	7.2 vehicles		
Residual risk		27%	
Operational disruption		LOW	

Intervention setup

Modelled from the current snapshot

RESPONSE PATH

Continue line

Accept current trajectory

Repair station

Reduce source risk

Targeted quarantine

Contain affected vehicles

Full line stop

Maximum risk reduction

TARGET STATION

ST-12 · Paint

Compare outcome

Projected outcome - ST-12

Repair Station

MEDIUM disruption

Expected queue	3.7 vehicles	Residual risk	1%
Queue impact	3.7 vehicles		
Residual risk		1%	
Operational disruption		MEDIUM	

Intervention setup

Modelled from the current snapshot

RESPONSE PATH

Continue line

Accept current trajectory

Repair station

Reduce source risk

Targeted quarantine

Contain affected vehicles

Full line stop

Maximum risk reduction

TARGET STATION

ST-12 · Paint

Compare outcome

Projected outcome - ST-12

Quarantine

LOW disruption

Expected queue	4.4 vehicles	Residual risk	1%
Queue impact	4.4 vehicles		
Residual risk		1%	
Operational disruption		LOW	

Intervention setup

Modelled from the current snapshot

RESPONSE PATH

Continue line

Accept current trajectory

Repair station

Reduce source risk

Targeted quarantine

Contain affected vehicles

Full line stop

Maximum risk reduction

TARGET STATION

ST-12 · Paint

Compare outcome

Projected outcome - ST-12

Full Line Stop

HIGH disruption

Expected queue	0.0 vehicles	Residual risk	1%
Queue impact	0.0 vehicles		
Residual risk		1%	
Operational disruption		HIGH	

WHAT-IF outcomes under different scenarios

HUMAN-IN-THE-LOOP : APPROVE, OVERRIDE, AUDIT

THE MODEL RECOMMENDS. A HUMAN DECIDES.

No autonomous action is ever taken on the physical line.

- **Every recommendation:** requires explicit Approve or Override
- **Decision notes:** optional free-text note captured with each action
- **Decision log:** timestamped, session-level audit trail
- **Accountability:** a person owns every decision made from this system

The interface is titled 'What-if simulator' with a subtitle 'Confidence-aware decisions for a 40-station vehicle assembly line.' It includes a 'SIMULATION SCENARIO' dropdown set to 'Hidden degradation' and a 'Run' button. A 'Human approval required' badge is present. The 'Intervention setup' section shows a 'RESPONSE PATH' with four options: 'Continue line' (selected), 'Repair station', 'Targeted quarantine', and 'Full line stop'. The 'TARGET STATION' is set to 'ST-12: Paint'. The 'Compare outcome' button is at the bottom. The right panel shows 'Projected outcome: ST-12' with 'Expected queue' at 3.7 vehicles, 'Residual risk' at 1%, and 'Operational disruption' at MEDIUM. The 'Human decision' section has 'Approve recommendation' and 'Override recommendation' buttons, with an 'Optional decision note' field. The 'Decision log' is empty.

The 'Continue line' option is now highlighted with a green checkmark. The 'Compare outcome' button is active. The right panel shows updated metrics: 'Expected queue' at 7.2 vehicles, 'Residual risk' at 27%, and 'Operational disruption' at LOW. The 'Human decision' section shows the 'Approve recommendation' button with a green checkmark and the note 'Continue production irrespective of defects'. The 'Decision log' now contains one entry: 'Approved - Continue line' at 'ST-12 · just now'.

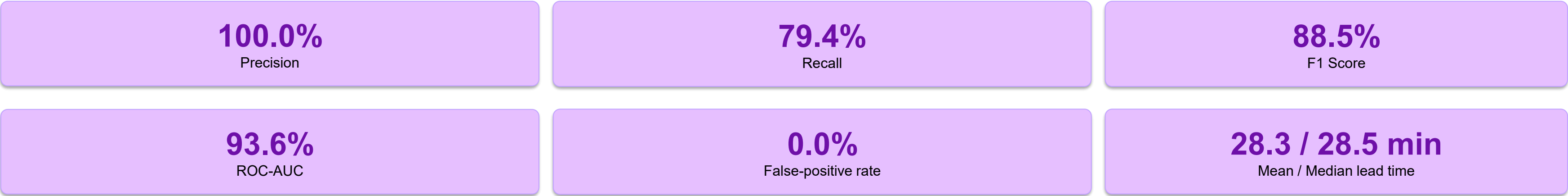
eg. Instruction worked after we compared outcome for the continue line option and can be seen in logs now

The 'Override recommendation' button is now highlighted with a red checkmark. The 'Compare outcome' button is active. The right panel shows the same metrics as the previous state. The 'Human decision' section shows the 'Override recommendation' button with a red checkmark and the note 'Stop Production'. The 'Decision log' now contains two entries: 'Overridden - Continue line' at 'ST-12 · just now' and 'Approved - Continue line' at 'ST-12 · just now'.

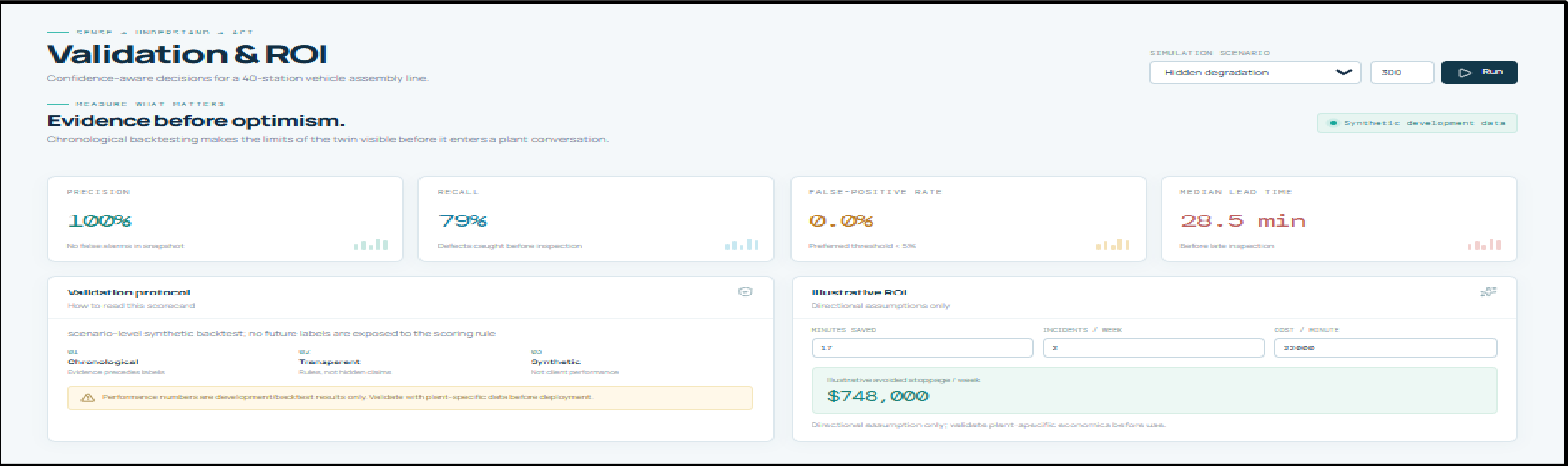
Here we overrode our previous instruction which is clearly seen in the logs

VALIDATION : EVIDENCE BEFORE OPTIMISM

SYNTHETIC SCENARIO-LEVEL BACKTEST RESULTS



These are synthetic scenario-level backtest results and are NOT claims of factory performance. Validate on plant-specific data before deployment.



STRESS-TESTED ACROSS SCENARIOS

FOUR CONDITIONS, ONE CONSISTENT ENGINE

Same detection and genealogy logic applied consistently across every scenario.

- Normal baseline
- Hidden degradation
- Bottleneck growth
- Sensor gap

SIMULATION SCENARIO

Hidden degradation

Hidden degradation

Bottleneck growth

Sensor gap

Normal baseline

300

Run

thetic development data

Note : All these four conditions are available across all the other pages so we can see the parameters regarding all these conditions on every page

LIMITATIONS & RESPONSIBLE POSITIONING

WHAT THIS PROTOTYPE DOES NOT CLAIM

We'd rather judges trust us because we're upfront about limits than oversell what this does today.

SYNTHETIC DATA

Built and validated on synthetic data, not live factory data.

RANKED, NOT PROVEN

Genealogy shows ranked evidence, not proof of physical causality.

RULE-BASED MODEL

Statistics- and rule-based today, not a trained production ML system.

NO LIVE OT LINK

No live PLC / OT / historian integration yet.

HUMAN IN THE LOOP

Every decision requires human approval - by design, not limitation.

PROTOTYPE DEMONSTRATION

Demonstration Video drive link :

<https://drive.google.com/file/d/1CDOYzyU7LZsiJgFwsgLqLXY25FHtkUbD/view?usp=sharing>

Working Prototype : <https://twin-line-ai-operations-twin--parthsbhagwat12.replit.app>

Thank you

